WORD-LEVEL SIGN LANGUAGE RECOGNITION USING SPATIAL–TEMPORAL GRAPH CONVOLUTIONAL NETWORKS (ST-GCN)

Students: Đặng Xuân Minh Hiếu (24002005), Mai Hoàng Đức (22001565), Lê Quý Công (22001550)

Supervisor: Dr. Cao Văn Chung

Faculty of Mathematics, Mechanics and Informatics
VNU University of Science

Hanoi, 2026

Outline

Motivation and Challenges

- **Significance:** Communication barriers faced by the deaf and hard-of-hearing community.
- **RGB challenges:** Background noise, viewpoint variation, and high computational cost.
- **Skeleton-based solution:** Noise-robust, lightweight, and geometrically stable.

Feature Extraction with the MediaPipe Tasks API

- Employs the **Pose & Hand Landmarker**.
- Extracts $V = 75$ landmarks:
  - 33 pose landmarks.
  - 21 left-hand landmarks.
  - 21 right-hand landmarks.
- Input tensor: $X \in \mathbb{R}^{3\times T\times 75}$.

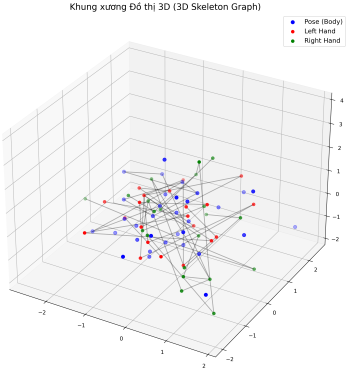
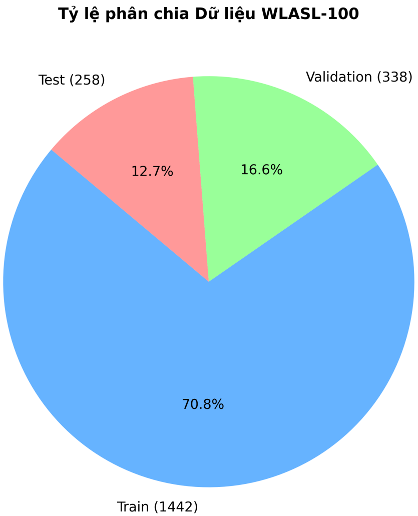


Figure: Skeletal connectivity structure

Analysis of the WLASL-100 Dataset



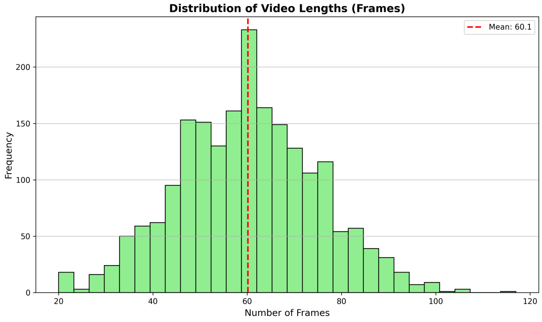


Figure: Sequence-length distribution

Improved ST-GCN Architecture

- **Depth:** Scaled up to 6 ST-GCN blocks.
- **Adaptive graph:** A learnable, dynamic adjacency matrix.
- **Pipeline:**
  1. Spatial GCN: learns hand shape.
  2. Temporal CNN: learns motion (kernel $= 9$).
- **Techniques:** Residual connections, batch normalization, dropout.

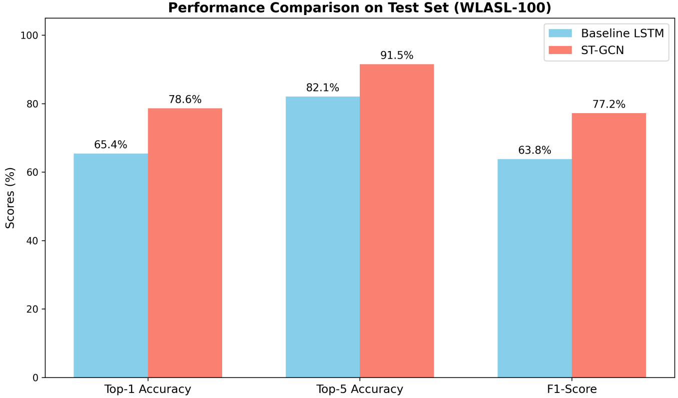


Figure: Accuracy on the test set

Measured ST-GCN metrics:

- **Top-1 accuracy:** 67.79%
- **Top-5 accuracy:** 88.0%

The model attains stable accuracy and recognizes the 100-word vocabulary well.

Confusion Matrix

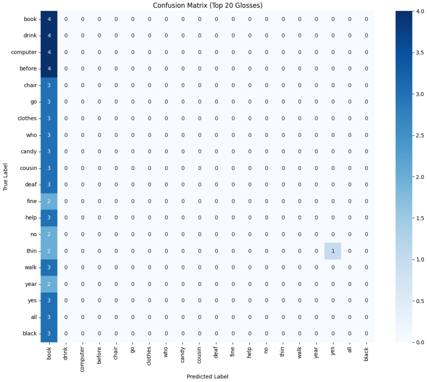


Figure: Confusion matrix (100 classes)

Conclusion

- Successfully built a real-time WSLR system.
- ST-GCN with an adaptive graph delivers superior performance.
- A Top-5 accuracy of 88.0% indicates readiness for practical deployment.

From State-of-the-Art Architectures to Low-Latency Systems Optimization

(Reference material for in-depth discussion)

The Core Bottleneck of ST-GCN

Limitations of a Static Adjacency Matrix

ST-GCN fixes the graph structure according to **human anatomy**.

- Connections between joints are hard-coded in advance.
- It cannot learn **flexible cross-links** among the hands, face, and body.
- Semantic cues are lost when a sign requires coordinating anatomically distant parts.

$\Rightarrow$ *Dynamic and multi-modal architectures are required to reach SOTA.*

Four Directions Beyond ST-GCN

1. **Multi-cue fusion** – combining multiple data streams.
2. **VAC** – a temporal alignment constraint.
3. **Gloss-free (VLM)** – learning without intermediate labels.
4. **CTR-GCN** – dynamic graphs per feature channel.

The following slides examine each direction in turn.

Direction 1 – Multi-cue Fusion

Ingesting Multiple Streams in Parallel

- **Skeleton:** hand shape and trajectory.
- **Raw RGB:** mouth shape and facial-expression cues.
- **Optical flow:** fine-grained motion direction.

- Each stream compensates for the blind spots of the others.
- The key issue is **how the streams are fused** (see next slide).

Direction 1 – The Cross-Attention Mechanism

Late vs. Early/Intermediate Fusion

Modern models fuse *early/intermediately* via cross-attention, rather than merging at the final layer (late fusion).

Attention formula

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

- $Q \leftarrow$ the **RGB** stream
- $K, V \leftarrow$ the **pose** stream

The model learns autonomously "when to attend to the face and when to attend to the fingers".

Direction 2 – VAC (Visual Alignment Constraint)

Weakness of the Conventional CTC Loss

It maps a 100-frame video to 3 glosses without specifying which gloss occurs at which frame $\Rightarrow$ the network becomes "lazy" and yields blurred features.

How VAC Addresses This

- Forces the model to produce a sharper **pseudo-alignment**.
- Encourages features of adjacent frames to cluster into distinct semantic groups.
- Adds auxiliary loss terms that assist backpropagation.

Direction 3 – Gloss-Free via VLMs

Joint Embedding Space (CLIP-style)

Contrastive learning pulls the embedding of a *signing video* close to that of its corresponding *text sentence* in a high-dimensional space.

Benefits

- Eliminates the intermediate gloss-annotation stage entirely.
- Saves annotation costs of up to millions of dollars on large datasets.

Direction 4 – Dynamic Graphs (CTR-GCN)

From a Static to a Per-Channel Dynamic Adjacency

Instead of a single anatomically fixed adjacency matrix A, CTR-GCN **learns parameters** that generate distinct graph connections for *each feature channel*.

- Each channel may carry a different graph topology.
- The left hand may **connect directly** to the right eye if that link is semantically meaningful.

The Time-Budget Constraint (30 FPS)

The Physical Law of Real-Time Processing

At 30 FPS, each frame has only 33.3 ms. Exceeding this budget $\Rightarrow$ frame drops and loss of motion data.

All five pipeline stages must fit within the 33.3 *ms* budget.

Challenge 1 – Transformer Cache Locality

Self-Attention Has $\mathcal{O}(N^2)$ Complexity

A long sliding window ($\sim$200 frames) produces enormous matrices.

- Matrices repeatedly overflow the L1/L2 cache (cache thrashing).
- Data is evicted to VRAM, or worse, to system RAM.
- Access speed can degrade by a factor of thousands.

Challenge 2 – Input Bandwidth Optimization

The Conventional Copy Path (Slow)

Camera $\rightarrow$ Kernel Space $\rightarrow$ User Space $\rightarrow$ PCIe $\rightarrow$ VRAM.

Kernel Bypass / Zero-Copy (GPUDirect)

- Writes the video stream **directly** into GPU VRAM.
- Bypasses the CPU and operating system entirely.
- Maximizes throughput, achieving microsecond-level latency.

Lesson 1 – The Smart-Glove “Graveyard”

Why Did Smart Gloves Fail?

Wearable-sensor solutions suffer from numerous physical limitations.

- Flex sensors are limited in their degrees of freedom (DoF).
- IMUs suffer from sensor drift when one hand occludes the other.
- They **ignore** the upper body and facial expression entirely – which carry substantial meaning.

Lesson 2 – The Core Infrastructure: Google MediaPipe

A Graph-Based Architecture Behind ∼90% of Real-Time Applications

The computation pipeline is divided into independent “nodes”.

- The hand-detection and eye-tracking nodes run independently and multi-threaded.
- They share **pre-allocated** tensors.
- This avoids garbage-collection latency within the real-time loop.

Lesson 3 – Domain Shift: B2C vs. B2B

Mobile Applications (B2C)

- Shaky camera angles and backlighting.
- Complex backgrounds corrupt the feature vectors.
- Prone to failure due to domain shift.

Fixed Stations (B2B) – SignAll

- Installed in schools and hospitals.
- Fixed cameras and standardized lighting.
- Reduces environmental error.

Why Abandon Python in Production?

Limitations of Native Python

- The **GIL** (Global Interpreter Lock) prevents true multi-threading.
- Large overhead from dynamic type checking.

Solution: ONNX + C/C++ with Memory Pooling

- Avoid calling `malloc`/`new` inside the inference loop (which causes tail latency).
- Allocate one large RAM block at start-up and reuse it for every frame.

The Three Graph-Optimization Levels of ONNX Runtime

Level 1 – Constant Folding

Pre-computes operations between constants once at load time $\Rightarrow$ fewer nodes to process.

Level 2 – Layer Fusion

Fuses Conv $\rightarrow$ BN $\rightarrow$ ReLU into a single **megakernel**; data stays in registers and is written to VRAM only once.

Level 3 – Hardware Execution Provider

Recompiles using native hardware APIs, e.g. **DirectML** (DirectX 12) for any GPU.

The Mathematics of Layer Fusion

Before fusion:

$$\text{Conv:} \quad y = Wx + b$$

$$\text{BN:} \quad z = \gamma \frac{y - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta$$

After fusion (re-parameterized weights):

$$W_{\text{fused}} = \frac{\gamma \cdot W}{\sqrt{\sigma^2 + \epsilon}}$$

$$b_{\text{fused}} = \frac{\gamma \cdot (b - \mu)}{\sqrt{\sigma^2 + \epsilon}} + \beta$$

Idea

A linear-algebraic transformation performed *prior* to inference.

Consequence

The BN layer is removed from the inference graph; millions of FLOPs are eliminated with error below 0.001%.

Summary of the Extended Material

- **Research:** Multi-cue + cross-attention, VAC, gloss-free (VLM), and CTR-GCN move beyond ST-GCN.
- **Systems:** The 33.3 *ms* budget, Transformer cache locality, and kernel bypass determine real-time feasibility.
- **Deployment:** ONNX Runtime + C++ (memory pooling, layer fusion) turn SOTA papers into products running smoothly at 30 FPS.

Thank you for your attention!